

Арифметика значений синуса и косинуса

Домашнее задание

1 сентября 2026 года

Теоретический минимум. Algebraic integers and minimal polynomials; $\deg_{\mathbb{Q}} \cos \frac{2\pi}{n} = \varphi(n)/2$; Niven's theorem for $\sin$, $\cos$ and $\tan$; the Chebyshev recurrence as an integrality machine; the classical irrationality proofs of π and π^2 ; the statement of the Lindemann–Weierstrass theorem.

1. Prove Niven's theorem for the tangent: if θ is a rational multiple of π and $\tan \theta \in \mathbb{Q}$, then $\tan \theta \in \{0, \pm 1\}$ (or is undefined).
2. (Niven 1947) Prove that π^2 is irrational by studying $I_n = \frac{1}{n!} \int_0^\pi x^n (\pi - x)^n \sin x \, dx$, and explain why the same argument gives the irrationality of π .
3. (Proposed problems) For $n \geq 3$ determine, effectively, the isomorphism class of the Galois group $\text{Gal}(\mathbb{Q}(\cos \frac{2\pi}{n})/\mathbb{Q})$.
4. Prove that $\cos 1^\circ$ is algebraic of degree 48 over $\mathbb{Q}$, while $\cos 1$ (one radian) is transcendental.
5. Describe all points of the unit circle with both coordinates rational through the substitution $t = \tan \frac{\theta}{2}$, and deduce the classification of Pythagorean triples.
6. Solve the equation $x^4 + x^3 + x^2 + x + 1 = 0$ by the substitution $y = x + \frac{1}{x} = 2 \cos \theta$, and deduce the classical values $\cos \frac{2\pi}{5} = \frac{\sqrt{5}-1}{4}$ and $\cos \frac{\pi}{5} = \frac{\sqrt{5}+1}{4}$. Do the same for $n = 7$ and explain why the answer there cannot be written in real radicals.
7. Let $a_n = 2 \cos(2^n \alpha)$ with $2 \cos \alpha = a_0 \in \mathbb{Z}$. Prove that all a_n are integers, and describe those α for which the sequence is periodic.

8. Prove that $\sum_{k=1}^{n-1} \cot^2 \frac{k\pi}{n}$ is an integer for every n , and give a proof that does not compute the sum explicitly.
9. (гауссовы периоды) For an odd prime p let $\eta = \sum_k \cos \frac{2\pi k}{p}$, the sum taken over the quadratic residues k modulo p . Prove that $2\eta + 1 = \pm\sqrt{p}$ when $p \equiv 1 \pmod{4}$, whereas $\eta = -\frac{1}{2}$ when $p \equiv 3 \pmod{4}$ (there the $\sqrt{p}$ hides in the imaginary part of the Gauss sum); check the answer for $p = 5, 7, 13$.
10. (Lehmer) Prove that $2 \cos \frac{2\pi}{n}$ generates the ring of integers of the real cyclotomic field, i.e. $\mathbb{Z}[2 \cos \frac{2\pi}{n}]$ is integrally closed; check the statement by hand for $n = 5, 7, 9$.